

Paul Abrams

Toronto, Ontario

Senior Full-Stack Developer · .NET & React/TypeScript · Microsoft Azure · Dynamics 365 · DevOps · 15+ Years Government / Enterprise

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Professional Summary

Senior full-stack developer with **15+ years** shipping web applications end-to-end in **complex enterprise and Government of Canada** environments — Department of Justice, IRCC, ESDC. Deep on **.NET / C#** server-side (n-tier ASP.NET serving 5,000+ users) and **React + TypeScript** on the front end (custom Azure DevOps web extensions, chess training app on the Lichess REST API, an end-to-end multiplayer web game shipped solo). **Microsoft Azure** fluency across PaaS — **Azure Functions, Azure SQL, Azure Storage, API Management, Key Vault, Entra/Azure AD** — and **event-driven / serverless** architectures. **Microsoft Dynamics CRM** delivery experience: engineered plug-ins, custom actions, and workflow processes in C# as part of the iCase → Dynamics modernization at DoJ, and SSIS ETL between legacy and replacement systems. **CI/CD** in **Azure DevOps** and **GitHub Actions** (Docker, Kubernetes, ITSG-33 / PBMM controls); **automated testing** (unit, integration, Selenium UI). **Professional Scrum Master I**. Active Agile delivery — sprint planning, stand-ups, retrospectives, cross-functional collaboration with BAs, QA, designers. Currently operating as an independent contractor; available within 1 week and open to relocation to Edmonton on contract award.

Professional Experience

Abrams Software Kingston / Toronto, Ontario

August 2025 – Present

Full-Stack Developer · Independent

- **TWNR** — **architected and shipped a multiplayer trading game as a full-stack web application, solo**. **React**-style component patterns on the front end (TypeScript); **Express + WebSocket** REST/event server on the back; **PostgreSQL** persistence with concurrency-correct transactions (`SELECT ... FOR UPDATE`); **JWT auth**; **JSON-Schema-generated REST API contract** that doubles as live documentation. Live at twnr.paulabrams.ca.
- **Chess Attack Visualization** — **React + TypeScript** training app consuming the **Lichess.org REST API**. Diagnosed a missing x-ray attacker feature in `chess.js` via React-lifecycle breakpoint debugging, opened [upstream PR #572](#), and published a fork as [@pabrams/chess.js](#) on npm. Live at paulabrams.ca/chess-attack-visualization.
- **LLM code evaluation** for DataAnnotation.tech and Stellar.ai — designing prompt fixtures and grading rubrics that probe specific model failure modes across **.NET / C#, TypeScript, React, Ruby, Python, Go, Rust, and SQL**, including security audits at known-vulnerable revisions with CWE/CVSS-graded inventories, adversarial unit tests, and AI-assistant “skills” evaluated through controlled ablation.
- **60+ public PRs** to outside repos — Microsoft’s `vsts-team-calendar` and `tfs-cli`, the official TypeScript-Website, the **Microsoft Authentication Library .NET** docs, `canada-ca/platform-security`, the Dygma split-keyboard configurator, and others.

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Software Engineer, Infrastructure and Cloud Enablement (BDM Project)

- Enhanced and maintained a **microservices** enterprise onboarding application — **C# / .NET, Blazor, Azure SQL, Azure Functions, GraphQL**. Designed and consumed **REST APIs** between services; integrated with enterprise IAM via **Azure AD / Entra (OAuth 2.0, OpenID Connect)**; published and governed APIs through **Azure API Management** with authentication, throttling, and transformation policies.
- **Event-driven workflows** via **Azure Functions** — Blazor front-end submissions kicked off Functions that triggered ADO pipelines, applied ADO security and work-item updates, integrated with on-prem

Active Directory via AAD Connect, and applied AAD group memberships based on personnel profile. Bulk submissions fanned out through asynchronous handoffs across the microservices.

- **Migrated the BDM Onboarding application's downstream identity** from rotating secrets (Key Vault entries + Azure DevOps PATs requiring 90-day rotation, with all auto-generated ADO work items attributed to a single person's identity) to **Azure Managed Identities** — service principals issued at runtime through Entra. Eliminated the secret-rotation incident class and properly attributed automated activity. Partnered with the Azure DevOps Services and Entra administrators on permission scoping.
- Designed and shipped **custom Azure DevOps web extensions in React, TypeScript, and Node.js** — a build/deploy analytics widget rendering success-rate charts via the ADO Builds REST API and Charts.js; a people-by-project reporter using an in-memory hashmap for tractable cross-referencing (the naive query pattern was unusably slow — a **performance** and **query-design** call); and a weekly work-item digest emailer driven by scheduled pipelines. Responsive UI, custom styling, accessibility-aligned interactions.
- Built **PowerShell tooling** against the Azure DevOps REST API to snapshot area paths, iteration paths, security configurations, and org-level state on a schedule and commit those snapshots into Git as a **configuration-drift detector** — automation that closed a gap no off-the-shelf product covered.
- **CI/CD pipelines** in Azure DevOps for build, test, **Docker** containerization, **Kubernetes** deployment, and **Azure Data Factory** (ETL) orchestration. Implemented **ITSG-33 (NIST-based) security controls** — TLS / SSL certificate management, secret hygiene via **Azure Key Vault** — for SA&A submissions against a **PBMM** security profile. Daily **Git** workflow.
- Ran weekly **demos and design walkthroughs** for junior developers on Azure DevOps, Key Vault, app registration, and custom extensions. Active participant in **daily stand-ups** with BAs, QA, and developers.

Employment and Social Development, Canada Gatineau, QC August 2019 – November 2021

Development Lead, Testing Services

- Co-architected an **internal C# / .NET web application** enabling QA staff and business analysts to author structured automated test scenarios via a web UI; designed the **SQL Server** schema and **co-designed the REST API on the web-app side** enabling bidirectional integration with Azure DevOps pipelines — script-fetch endpoints (versioned Selenium scripts generated on the fly from SQL) and results-push endpoints for structured test outcomes. Integrated with the departmental headless CMS platform.
- Engineered Azure DevOps **CI/CD pipelines** for **automated testing** — provisioned Windows agents to execute parallel **Selenium** UI tests against deployed applications and integrated logged test results back into ADO work items.
- Championed the departmental transition from **LoadRunner to JMeter**, reducing licensing costs while maintaining robust performance-testing capabilities. Code review and standards enforcement for the team.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead

- Led **Agile / Scrum** sprint planning, review, retrospective, **daily stand-up**, and backlog refinement; pioneered capacity-planning and forecasting using Azure DevOps Server; introduced poker planning estimation across a cross-functional team.
- **Mentored developers** in **SQL Server query optimization** — query tuning, indexing strategy, partitioned tables, execution-plan analysis — against large TFS databases and warehouses. Also coached on SSRS and C# / .NET.
- Implemented and deployed **custom web extensions in React / TypeScript / Node.js** with responsive layouts and accessibility-aligned interactions.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Development Lead — iCase / Microsoft Dynamics CRM Modernization

- Dev lead on the **Legal Case Management System (iCase)** — a large n-tier **ASP.NET / C#** enterprise web application used by 5,000+ lawyers — while concurrently building its **Microsoft Dynamics CRM** successor. Modified iCase schema and front-end UI to surface migration status to users in real time, and adjusted the integration stored procedures (icisp_*) consumed by the department's SAP-based financial information system.
- Engineered **MS Dynamics CRM plug-ins, custom actions, and workflow processes** in **C#**; rewrote SQL stored procedures as C# with LINQ inside CRM; used **SSIS** for ETL between iCase and Dynamics. Applied design patterns (Repository, Strategy, Mediator) and clean-code refactoring. Deployments to Windows servers via both manual runbooks and scripted pipelines.
- Established **code-review processes** integrated directly with TFS work items — gating standards for maintainability, security, and performance — and mentored junior developers.

Department of Citizenship and Immigration, Canada Ottawa, ON April 2014 – December 2015

Senior Software Engineer

- Hired alongside one other engineer to take over a **TFS / TFVC** release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop release-management apps** and replaced them with lightweight PowerShell scripts calling the **TFS REST APIs** — net result, most of the contractor tooling was simply deleted. **Everything about the application ended up in source control** — group memberships, permissions, repository access, process template customizations. Bias for action and ownership.
- Implemented and deployed Team Foundation Server **web extensions** in **React / TypeScript / Node.js**; maintained a fork of Microsoft's vsts-team-calendar extension with configurable colours, Outlook integration, release-management features, and IE-compatibility shims.
- Guided development teams in migrating from legacy XAML builds to modern, scriptable build definitions; deployed across Windows environments both manually and via automated pipelines.

Department of Justice, Canada Ottawa, ON October 2008 – April 2014

Full-Stack Developer (2009–2014) · Software Development Engineer in Test (2008–2009)

Core contributor to the department's flagship **Legal Case Management System (iCase)** — a distributed multi-tier **ASP.NET / C#** enterprise web application used by 5,000+ lawyers and agents across Canada (including external Crown agents). Multi-tier with role- and object-based security, integrated with records management, financial, and email systems, 24/7 high availability.

Full-Stack Developer (2009–2014)

- Led design and largely solo implementation of the **"My iCase" widget portal** — the new entry point to iCase, serving thousands of internal and external users. Ran JAD sessions, produced prototypes, authored BDD/TDD documentation. Built on **ASP.NET Web Parts and the Personalization framework** with a **custom SQL personalization provider, Knockout.js MVVM** on the front end (a direct ancestor of the **React** patterns I use today), async AJAX web-service calls, async HTTP handlers, ReportViewer in remote SSRS mode, and an HTML5 polyfill for the department-mandated IE8 environment.
- Diagnosed a homepage report taking ~10 s uncached: profiled with **SQL Profiler, Report Execution logs, and actual execution plans**, then replaced live joins with a flat table populated by a nightly job. **~50x faster uncached, 3-4x faster cached**. Textbook **query-tuning through execution-plan analysis**.
- Built a **self-initiated availability-monitoring suite** for web and document servers because support staff were running checks manually evenings and weekends; eliminated significant overtime and reduced operational risk.
- Diagnosed and resolved a **COM Interop memory leak** on the document servers; built automated health monitoring to catch regressions.
- Co-led the **SQL Server 2005 → 2012** database upgrade including dev coordination and testing; rebuilt iCase's UI as a widget-based Knockout.js interface to eliminate full-page postbacks; contributed to the

multi-year migration from **Crystal Reports to SSRS**, authoring 14 new SSRS reports for the Legal Risk Management module.

Software Development Engineer in Test (2008–2009)

- Identified a critical regression-testing bottleneck and voluntarily architected a custom framework in **IBM Rational Functional Tester** that parsed English-language scenario documents and auto-generated executable test scripts.
- Replaced the department's **"all-hands" manual regression cycle** (~20–25 testers running for two weeks every release) with one person reviewing automated results in five days. **~98% reduction in release validation effort**, eliminated the standing 2-week code freeze per release, **saved an estimated \$160,000 per cycle**.

Education

Bachelor of Computer Science (Honours) - Carleton University

September 1999 - May 2003

Certifications & Awards

[Professional Scrum Master I](#) (Scrum.org) · [ITIL v3 Foundation](#) (Loyalist Certification Services) · DOJ Team Merit Award 2012 (Cost Recovery Process Improvement); Team Merit Award 2011 (Chart of Accounts); Team Spirit Award 2009